

# The General Gauss–Markov Model

Every chapter so far has assumed that the errors are uncorrelated and equally variable. That assumption is a statement about the *geometry* of observation space: it says that Euclidean distance is the right way to measure how far the data are from the model. When observations share a common shock, follow one another in time, or are measured with different precision, Euclidean distance is the wrong ruler. This chapter replaces  $\sigma^2\mathbf{I}$  by a general  $\sigma^2\mathbf{V}$ : first with  $\mathbf{V}$  known and invertible, where the theory of Parts II and III survives a change of coordinates; then in the two places where it does not survive, when  $\mathbf{V}$  is singular and some functions of  $\boldsymbol{\beta}$  become known exactly, and when  $\mathbf{V}$  has to be estimated from the data that produced the fit.

**What you need.** Chapter 6, especially Section 6.9: projection in the  $\mathbf{V}^{-1}$  inner product (Definition 6.9.1, Theorem 6.9.1), oblique projections (Proposition 6.9.4) and Kruskal’s theorem (Theorem 6.9.3). Section 7.2 for the Gauss–Markov theorem (Theorem 7.2.1) and Aitken’s corollary (Corollary 7.2.3), which this chapter cites and does not reprove. From Chapter 1: generalized inverses, square roots (Theorem 1.7.6) and the Woodbury identity (Theorem 1.6.5). From Chapter 2: whitening (Proposition 2.2.4) and the support of a singular covariance (Theorem 2.2.3). Estimability (Chapter 8) and the  $F$  test (Chapter 11) are used throughout, and the distribution theory of Section 31.1 rests on the quadratic forms of Chapter 4. Section 19.2 and Chapter 21 supply the diagnostic side: what a wrong covariance does (Theorem 19.2.2), how much efficiency it costs (Theorem 19.2.3), and the sandwich estimator (Theorem 21.4.2).

## 31.-1.1 Roadmap

- Generalized least squares
- When ordinary least squares is best
- Singular covariance matrices
- Estimated covariance: feasible generalized least squares
- Summary and notes